We chose Hiredly, a Malaysian job-search platform run by Agensi Pekerjaan Wobb Sdn Bhd, a licensed employment agency (JTKSM 1033). It has a website and iOS and Android apps, with over 100,000 Android downloads. Job seekers use it for free, while employers pay for job credits, listing upgrades and access to a resume database. It also sells a headhunting service, Hiredly X, and a set of AI features: job matching, a job-description summariser and a CV builder.

Hiredly meets all three conditions in the brief. It is real and available, since every member of our group could install it and create an account. It is local and niche. It mainly serves Malaysia, and it is much smaller than regional platforms like JobStreet or global ones like LinkedIn. When we searched Google Scholar and university databases, we found no academic papers on Hiredly itself, only news articles and company material. Finally, it has clear ethically relevant features. It collects personal and demographic data, makes automated decisions that shape who sees which job, and earns its money from one side of the market while serving the other.

Our reasons for choosing it are also personal. One member has used Hiredly for two years while job hunting in Malaysia, and the rest of us will enter the same job market soon. A job platform sits at a sensitive point in people's lives, when they are under pressure, short of money and likely to accept whatever terms they are shown. That made it feel more important to us than a shopping or entertainment app. It is also the kind of system many of us could end up building as software engineers, so the questions we ask about it are ones we may have to answer at work.

Preliminary Investigation

We combined four methods so that no finding rests on a single source (1) Hands-on walkthrough: Each of us signed up, built a profile, uploaded a CV, searched for jobs and tried the AI features on both the website and the app. We noted every point where the platform asked for data, made a choice for us, or showed AI-generated content. (2) Policy analysis: We read the full Privacy Policy (dated 29 March 2024) and the Terms and Conditions, and recorded the clauses about data collection, sharing, consent, retention and liability. We then compared what the policy says with what we saw in the app. (3) Store label audit: We compared the data-handling declarations on the Google Play Data safety section and Apple's App Privacy label with each other and with the Privacy Policy. (4) Pricing and interface review: We looked at what employers can buy, including job credit plans and per-post upgrades, and checked whether paid listings are marked as paid to job seekers. We also searched for listings with badges and urgency prompts.

Ethical issues identified

Issue 1: Race and ethnicity data in opaque AI matching. The Privacy Policy lists "race or ethnicity" among the demographic data Hiredly collects, even from anonymous visitors, yet it also states "We do not process any sensitive data". Registration data feeds a "profiling engine" and "AI-powered job matching", and employers can use a "Digital Headhunter" that scans 1.3 million profiles to contact the top 1,000. Nothing tells users which signals drive the matching or why they were or were not shown. In Malaysia, where hiring discrimination by race is well documented, this is a serious fairness risk.

Issue 2: Consent to CV use is assumed, not asked. Under Terms clause 11.1, a stored CV is "automatically searchable" by employers who pay. The same clause admits Hiredly "cannot guarantee that other parties will not gain access", and employers may keep copies after the user deletes it. The policy treats consent as given "by continuing to access the Site", allows sharing with "data aggregators, marketers or others", and says users "should not expect" their data to stay private. The Google Play Store label, meanwhile, says no data is shared with third parties, by which Hiredly should comply with but does not make effort to do so.

Issue 3: AI summaries and the CV builder may misrepresent information. The app summarises long job descriptions into highlights and turns a profile into an "ATS-friendly" CV in one click. In the app, users are not clearly told this content is AI-generated. The website shows an "AI-generated summary BETA" label, but it carries no warning that details like salary or requirements may be dropped or wrong. Users acting quickly on a summary may apply for, or skip, a job based on false information.

Issue 4: Paying employers get advantages that job seekers don't. All of Hiredly's revenue comes from employers. Paid upgrades such as Spotlight (RM100), Refresher (RM50) and Featured (RM800) push a post higher or keep it near the top. Paying employers also receive job seekers' CVs and contact details (Terms 11.3). We found no label telling job seekers which listings are paid. Listings are also not screened (Terms 14.1), and Hiredly caps its liability to users at RM60 (Terms 16.1).

Literature review

No research exists on Hiredly itself, so we reviewed work on comparable systems under four themes.

Issue 1: Field experiments establish the human baseline. Bertrand and Mullainathan (2004) showed callbacks depending on the race and name signals, and the Cent-GPS (2019) study found the same in Malaysia. Across 3,829 identical applications, a Chinese candidate needed about 2 applications per callback while an Indian candidate needed 27. Audit studies show algorithms can reproduce this without a race field. Chen et al. (2018) found resume search engines ranked men above women with no gender input, and Lambrecht and Tucker (2019) showed ad delivery alone sent a neutral job ad to fewer women. Reviews by Bogen and Rieke (2018) and Köchling and Wehner (2020) agree that most bias enters at sourcing and ranking, which is where Hiredly's matching sits. The sources disagree on the remedy. Raghavan et al. (2020) argue that auditing needs demographic data, while the privacy literature below argues against collecting it.

Issue 2: Researchers agree that click-through consent fails, but explain it differently. McDonald and Cranor (2008) treat it as a time cost, estimating about 200 hours a year to read every policy a person agrees to. Obar and Oeldorf-Hirsch (2020) show it is also behavioural, since most participants skipped a fake policy entirely. Solove (2013) goes further and argues that even an informed click is not meaningful, because people cannot foresee how their data will be combined later. Nissenbaum's (2004) contextual integrity gives the sharpest test for Hiredly. A CV shared "for a job I applied to" is moved out of context when it is sold to aggregators and marketers.

Khandelwal et al. (2024) add that app store privacy labels often contradict actual practice, as Hiredly's do.

Issue 3: Two literatures combine here. Generation research finds summaries are often unfaithful: Maynez et al. (2020) found most abstractive summaries contained unsupported content, and Ji et al. (2023) describe hallucination as a general property of these models. Automation bias research finds users rarely catch such errors. Parasuraman and Manzey (2010) and Goddard et al. (2012) show people check automated output less, especially under time pressure. The first literature looks for fixes in model accuracy, while the second says disclosure and interface design matter just as much. On CVs, Fuller et al. (2021) found automated screening filters out many qualified candidates, and Wilson and Caliskan (2024) found language-model screening favoured white-associated names in 85% of tests. Malaysia's AI governance guidelines (MOSTI, 2024) call for generative AI use to be disclosed, though they are not binding.

Issue 4: Rochet and Tirole (2003) explain why platforms charge one side and subsidise the other, and they treat free access as a benefit to the subsidised side. Ranking research challenges how benign this is. Ursu (2018) showed through a randomised experiment that position causally changes what people click, so paid placement changes which jobs seekers consider. Edelman (2011) and the FTC (2013) argue that paid results must be clearly labelled to avoid misleading users, and Gillespie (2018) argues that ranking is an editorial act, so platforms cannot claim neutrality.

Ethical framework proposal

Here we will judge each issue against the ACM/IEEE-CS Software Engineering Code.

Issue	Clauses	Why they apply
Race data and AI matching	8.07, 3.13, 1.03, 6.07	Race is an irrelevant factor for job fit (8.07). Demographic data from anonymous visitors is not clearly authorised for ranking (3.13). Hidden ranking affects income (1.03), and "AI-powered matching" is advertised without explanation (6.07).
Assumed consent	3.12, 1.03, 3.14, 1.06	"Searchable by default" is a design choice against privacy (3.12, 1.03). Soft deletion keeps outdated CVs circulating (3.14). The store label contradicts the policy (1.06).
AI summaries and CV builder	6.07, 1.06, 3.10, 3.01	"ATS-friendly" is unverifiable (6.07). Unlabelled AI text can pass as the employer's own words (1.06). Summaries are a known failure mode that needs testing (3.10), and the speed-versus-accuracy tradeoff is hidden from users (3.01).

Issue	Clauses	Why they apply
4. Paid employer advantages	1.02, 1.06, 3.01	Ranking decides whose interests win (1.02). Unlabelled paid placement misleads (1.06), and users are not told that rank reflects payment (3.01).

We propose two theories from the module, each suited to different issues.

Kantian Deontology fits Issues 1 to 3, which are about how people are treated rather than about overall outcomes. Collecting ethnicity while denying it, assuming consent to share CVs, and passing off generated text as the employers all use job seekers as a means and break the duty of truthfulness. Deontological lenses let us call these wrong even if they increase hiring efficiency. The lecture example of a privacy policy that "technically states" data is sold matches Issue 2.

Utilitarianism fits Issue 4, which is a genuine weighing of outcomes across groups. Paid ranking gives employers reach, keeps the service free for seekers and keeps Hiredly in business, but some seekers never see better-matched jobs. Utilitarianism lets us weigh these effects, though we expect to meet its ‘calculation’ problem, because the cost of a job never seen is hard to measure.

Issue 4 is our main case; Principle 2 tells engineers to serve their client and employer "consistent with the public interest". On Hiredly the paying employers are the client, while job seekers appear only as users under Principle 1. The Code gives no rule for weighing a paying client against a non-paying user in a two-sided market. The decision to sell ranking is also made by management, outside the engineer's control. So the Code can tell an engineer to raise concerns (2.07), but it cannot say whether the business model is acceptable, and a theory is needed for that. Issue 1 adds a second case of clauses in tension. Clause 8.07 suggests collecting race to audit for bias, while 3.12 and 3.13 suggest not collecting it at all, and the Code does not say which should win.

Our Plan for Part 2

Part 2 will turn these observations into measured evidence. For Issue 1, we will create matched test profiles that differ only in name and record whether "Jobs for you" results change. Because the Terms forbid false profile information, we will confirm this method with Dr Nabil first, and we will never apply to real jobs with these profiles. For Issue 2, we will make data access and deletion requests under the PDPA and log the response time, any fee and what is actually removed. For Issue 3, we will compare 10 AI summaries with their full job descriptions and count omissions and errors, especially salary and requirements. For Issue 4, we will run the same searches daily for a week and record whether badged or refreshed listings outrank newer or better-matched ones. We will also code a sample of app store reviews by theme. We will then apply the clauses and theories above to each finding and reach a judgement for each issue.

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